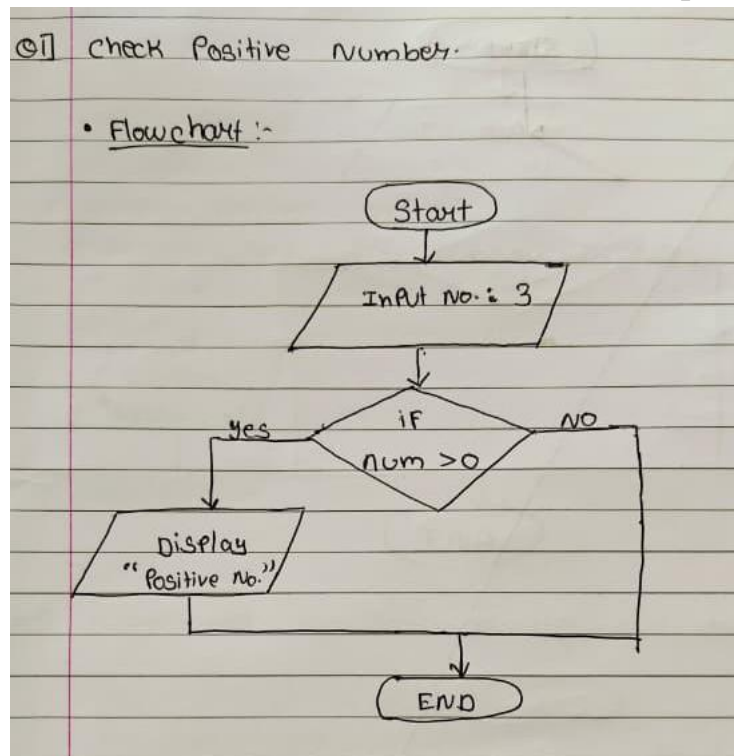


Assingment No.: 01

1. Check Positive Number:

- **Task:** Create a flowchart to check whether a number is positive.



- **Next Step:** Write a Java program that checks if a predefined number is positive using an if-else statement and prints the appropriate message.

Source code:

```
package LB_Assigment1;

public class Positive {
    public static void main(String[] args) {
        int num=3;
        if(num>0){
            System.out.println(num+" is Positive Number");
        }
    }
}
```

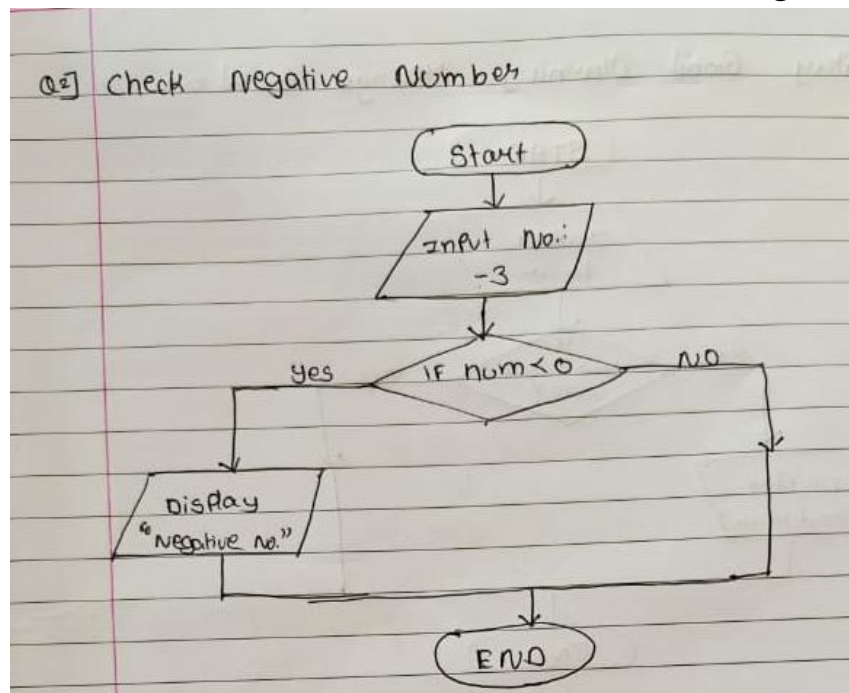
Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
3 is Positive Number

Process finished with exit code 0
```

2. Check Negative Number:

- **Task:** Create a flowchart to check whether a number is negative.



- **Next Step:** Write a Java program that checks if a predefined number is negative using an if-else statement and displays the result.

Source code:

```
package LB_Assignment1;
```

```
public class Negative {
    public static void main(String[] args) {
        int num = -3;
        if(num < 0){
            System.out.println(num + " is a Negative Number");
        }
    }
}
```

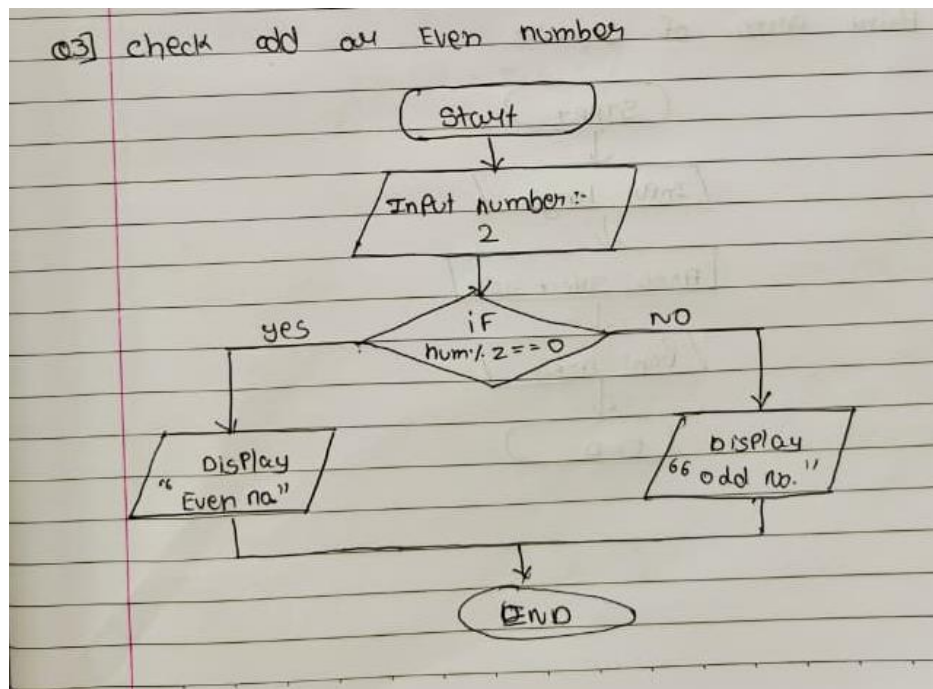
Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
-3 is a Negative Number

Process finished with exit code 0
```

3. Check Odd or Even Number:

- **Task:** Create a flowchart to determine whether a number is odd or even.



- **Next Step:** Write a Java program that checks if a predefined number is odd or even. Use an if-else statement and the modulus operator (%) to determine whether the number is divisible by 2 or not.

Source code:

```
package LB_Assignment1;

public class Even_Odd {
    public static void main(String[] args) {
        int num=3;
        if(num%2==0){
            System.out.println(num+" is a Even Number");
        }else{
            System.out.println(num+" is a odd Number");
        }
    }
}
```

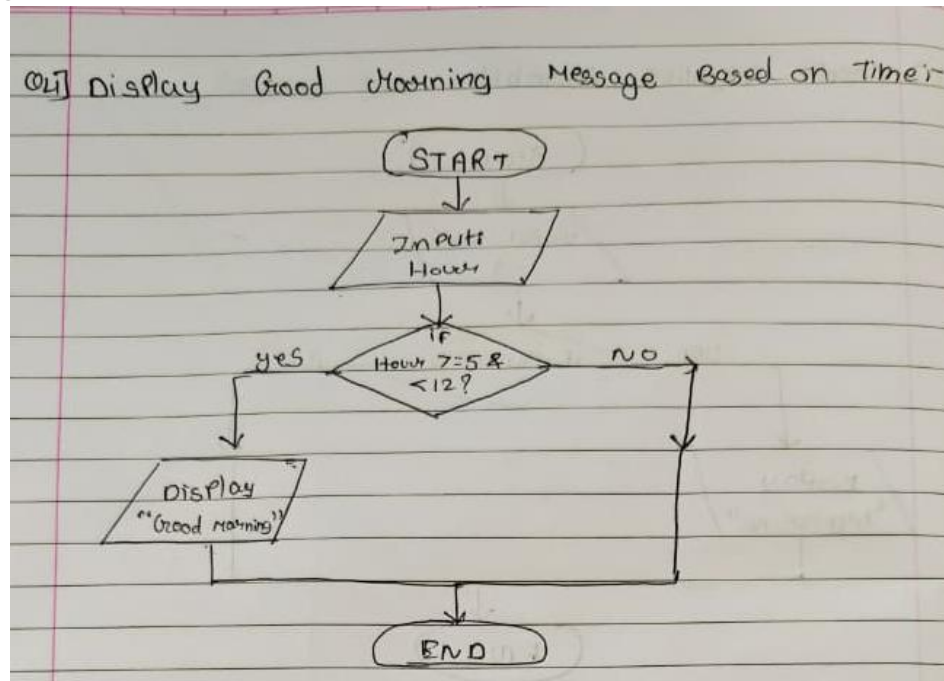
Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe" "
3 is a odd Number

Process finished with exit code 0
```

4. Display Good Morning Message Based on Time:

- **Task:** Create a flowchart to display a "Good Morning" message based on a given time.



- **Next Step:** Write a Java program that displays a "Good Morning" message if the predefined time is between 5 AM and 12 PM. Use an if statement to implement the logic.

Source:

```
package LB_Assignment1;
```

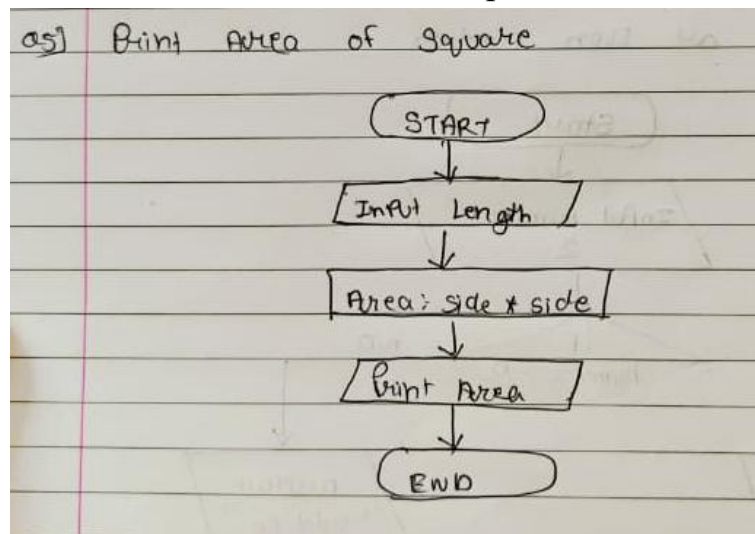
```
public class GM_BOT {  
    public static void main(String[] args) {  
        int time=10;  
        if(time>=5 && time<=12){  
            System.out.println("Good Morning");  
        }  
    }  
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"  
Good Morning  
  
Process finished with exit code 0
```

5. Print Area of a Square:

- **Task:** Create a flowchart to calculate and print the area of a square.



- **Next Step:** Write a Java program that calculates the area of a square using the formula $\text{area} = \text{side} * \text{side}$. Use a predefined side length.

Source code:

```
package LB_Assignment1;
```

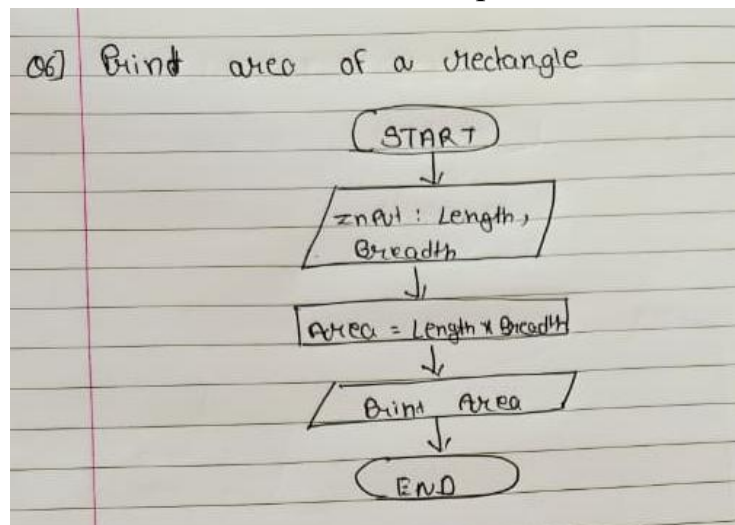
```
public class Area_of_Square {  
    public static void main(String[] args) {  
        int side=6;  
        int Area=side*side;  
        System.out.println("Area of Square of Side "+side+" is "+Area);  
    }  
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"  
Area of Square of Side 6 is 36  
  
Process finished with exit code 0
```

6. Print Area of a Rectangle:

- **Task:** Create a flowchart to calculate and print the area of a rectangle.



- Next Step: Write a Java program that calculates the area of a rectangle using the formula $\text{area} = \text{length} * \text{width}$. Use predefined values for length and width.

Source code:

```
package LB_Assignment1;
```

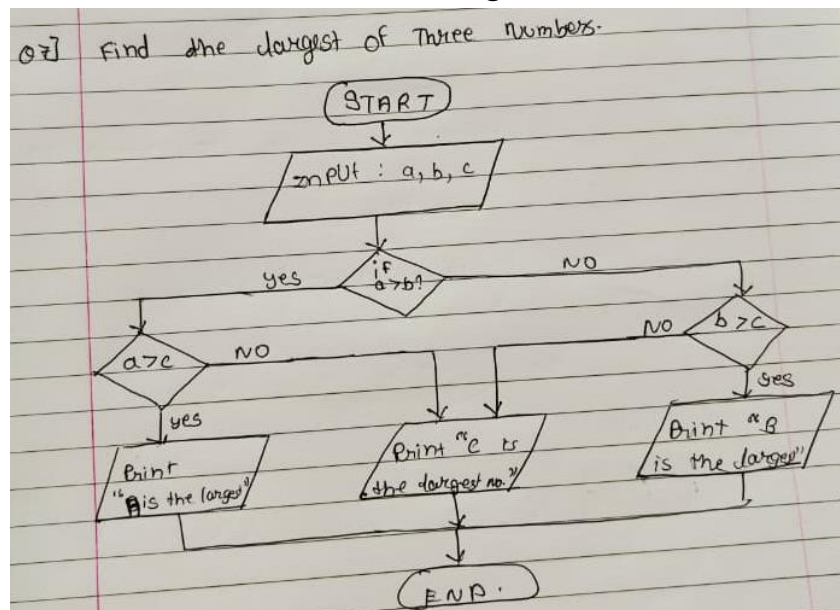
```
public class Area_of_Rect {  
    public static void main(String[] args) {  
        int Length=6;  
        int Breadth=3;  
        int Area=Length*Breadth;  
        System.out.println("Area of Rectangle of Length "+Length+" & Breadth  
"+Breadth+" is "+Area);  
    }  
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-javaagent  
Area of Rectangle of Length 6 & Breadth 3 is 18  
  
Process finished with exit code 0
```

7. Find the Largest of Three Numbers:

- **Task:** Create a flowchart to find the largest of three numbers.



- Next Step: Write a Java program that finds and prints the largest of three predefined numbers using if-else statements.

Source code:

```
package LB_Assignment1;
```

```
public class Largest_of_Three_nos {  
    public static void main(String[] args) {  
        int a=4;  
        int b=6;  
        int c=2;  
  
        if(a>b){  
            System.out.println("a = "+a+" is the Largest Number");  
        } else if (b>c){  
            System.out.println("b = "+b+" is the Largest Number");  
        } else {  
            System.out.println("c = "+c+" is the Largest Number");  
        }  
    }  
}
```

Output:

```
"C:\Program Files\Java\jdk-20\bin\java.exe"  
b = 6 is the Largest Number  
  
Process finished with exit code 0
```